

# Assignment 4

## Supervised vs Unsupervised Learning

### Part A – Explain in Your Own Words

#### 1. What is Supervised Learning?

Supervised Learning is a type of Machine Learning where the model is trained using labeled data. Each input has a correct output, and the model learns the relationship between them to make predictions on new data.

**Example:** Predicting house prices using historical house data and known prices.

#### 2. What is Unsupervised Learning?

Unsupervised Learning is a type of Machine Learning where the model is trained using unlabeled data. The model identifies hidden patterns, relationships, or groups within the data without being given correct answers.

**Example:** Grouping customers into different market segments based on purchasing behavior.

#### 3. Real-Life Analogies

##### **Supervised Learning:**

Studying with an answer key. You learn by comparing your answers with the correct solutions.

##### **Unsupervised Learning:**

Sorting photos into groups without any labels or instructions. You organize them based on similarities you observe.

#### 4. Difference Between Classification and Regression

Both Classification and Regression are types of Supervised Learning.

- **Classification** predicts categories or labels.
  - Example: Spam or Not Spam.
- **Regression** predicts continuous numerical values.
  - Example: Predicting a car's selling price.

In simple terms, Classification predicts a category, while Regression predicts a number.

## Part B – Classify Each Scenario

| Real-World Task                                                | Learning Type | Classification / Regression |
|----------------------------------------------------------------|---------------|-----------------------------|
| Predict tomorrow's temperature from weather data               | Supervised    | Regression                  |
| Group 10,000 customers into similar segments (no labels given) | Unsupervised  | N/A                         |
| Decide if an email is spam or not spam                         | Supervised    | Classification              |
| Predict the selling price of a used car                        | Supervised    | Regression                  |

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### Conclusion

Understanding the difference between Supervised and Unsupervised Learning is essential in Machine Learning. The presence of labeled data determines whether a problem is supervised or unsupervised, while supervised problems can further be categorized as Classification or Regression based on the type of output being predicted.